

A Behavior-Coordinate Realization Interface for Predictive Saturation Management

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Abstract—When a robot controller requests behavior its actuators cannot realize, torque clipping keeps the hardware command legal but silently changes the behavior executed: the robot is no longer running the PD, impedance, or learned law that was designed or trained, and nothing in the loop announces it. This matters directly in physical interaction, where the requested acceleration sets apparent compliance and disturbance response. We present a behavior-coordinate realization interface that detects and corrects this condition before clipping is reached. Heterogeneous nominal controllers expose a common task-acceleration request at servo rate; a slower manager reconstructs the executing robot’s configuration-dependent torque-feasible set, predicts loss of realizability over a short horizon, and returns a minimum-cost correction in the same behavior coordinates, with a fast projection retained as a last-resort actuator guard. The interface is designed to be auditable in deployment: three measurable conditions — torque-prediction accuracy, actuator margin, and behavior-successor mismatch — decide online whether the acceleration-to-torque interface can currently be trusted, and a one-step result establishes that the behavior certificate is preserved whenever they hold. A corollary gives the additional torque-drift and averaged-request conditions required by the two-rate implementation. A 111-run deterministic study evaluates the interface across five nominal-controller forms, three realization maps, eight stress scenarios, and a matched horizon trajectory-reference governor. Full-horizon enforcement removes a 3.587 Nm predicted future torque violation and 31.180 mm of workspace excess that first-step-only enforcement leaves behind. Under held-out plant perturbations the audit reports positive margins with maximum ℓ_∞ successor defects below 0.0069 m/s against a 0.03 m/s budget across all three realization maps. Against the matched governor the proposed vector correction attains lower correction RMSE in three of the four non-failure cases. The study also reports the conditions under which the interface fails: abrupt disturbance and severe model mismatch violate the audit premises for both predictive architectures, leaving only the reactive fallback. The 111-run study is deterministic and fully reproducible; Section VIII states where the interface applies.

Index Terms—Physical interaction, actuator saturation, behavior-coordinate interface, predictive constraint management, control refinement.

I. INTRODUCTION

A controller computes a torque request τ^0 and the hardware applies

$$\tau^{\text{app}} = \text{clip}(\tau^0, \tau_{\min}, \tau_{\max}),$$

that is, each joint’s torque is pushed back to its nearest limit whenever the request exceeds it. The clip protects the actuator. It does not protect the behavior. Once the clip is active, the

acceleration the robot produces is not the acceleration the controller asked for, so the robot is no longer running the PD, impedance, or learned law that was designed or trained. Nothing in the loop announces this. The controller keeps issuing requests, the actuators keep saturating, and the realized closed-loop dynamics silently drift away from the intended ones. In physical interaction this matters directly, because the requested acceleration is what sets apparent compliance and disturbance response.

Different controllers reach this state through different routes. A fixed-gain PD law gets there through a large tracking error. An impedance law $M_d\ddot{e} = -K_d\dot{e} - D_d\dot{e} + F_h$ gets there through contact force, even when tracking error is small. A tanh-squashed learned policy bounds its requested *acceleration*, but the torque that acceleration implies depends on the configuration, so a bound on the request is not a bound on the torque. An upstream language- or diffusion-conditioned module may propose a motion primitive with no representation of the executing robot’s actuator limits at all. The route differs; the failure is the same.

Keep the nominal controller unchanged, but require it to expose a task-acceleration request. Add a much slower layer that predicts whether each robot can realize that request and outputs a correction in the same behavior coordinates.

The nominal controller runs unchanged at its own high servo rate. A much slower manager rolls its behavior request forward over a short horizon, reconstructs the executing robot’s torque-feasible acceleration set, and, when its QP is feasible, applies a minimum-cost correction to the requested acceleration before hard clipping is required.

The manager does not choose the task and does not replace the controller. Because its job is to predict and correct rather than react, it also does not need to run at servo rate; how slow it can run in general depends on the operating region and model (a concrete rate pair is instantiated in Section VII). Its only job is to keep the request physically producible while staying as close as possible to what the controller asked for. A final high-rate projection stays in place for disturbances that arrive between manager updates; prediction and last-resort protection are separate responsibilities, and Section VII shows they can be needed at different moments.

A one-step check answers “is the command legal right now?” That is not the same question as “does this planned sequence remain legal?” Position and velocity carry forward, so a future constraint can require changing acceleration before the limiting step arrives. This distinction is standard in predictive constraint management [3], [5], [16]–[18]. Fig. 1

isolates its role inside the proposed interface: constraining every predicted move eliminates the planned torque excess in the horizon-ramp case, while constraining only the first move leaves a 3.587 Nm future violation and 31.180 mm of workspace excess (Section VII-B).

Feedback linearization to a double-integrator-like behavior model is classical [1], [2]. Reference governors, generalized action governors, predictive safety filters, anti-windup control, model-predictive constraint handling, and task-space capacity polytopes are established [3]–[7], [16]–[20]; the double integrator, the QP, and the ray-to-polytope authority calculation are used here as implementation ingredients. The contribution is their arrangement into an auditable interface, described next.

The question we do ask is this:

Can a common behavior-acceleration request be supervised without hiding robot-specific actuator geometry, and can the boundary between behavior preservation and physical realization be stated as quantities that an implementation can audit?

The proposed answer is an interface decomposition: behavior dynamics provide the shared request and certificate coordinates, while the torque map, feasible set, and uncertainty bounds remain robot-specific. Section VI specializes existing refinement logic [8], [9], [19] to three quantities that expose when this particular acceleration-to-torque interface can be trusted.

Contributions.

- 1) A behavior-coordinate realization interface in which heterogeneous nominal controllers expose the same task-acceleration request while every executing robot retains its own configuration-dependent acceleration-to-torque map and feasible set.
- 2) A two-rate implementation that corrects the behavior request predictively, recomputes the robot-specific realization map on the fast path, and reports both scalar torque headroom and directional acceleration authority; the components are assembled to preserve the interface across heterogeneous controllers and robots.
- 3) An actuator-realization specialization of one-step certificate refinement, separating torque-prediction accuracy (T1), actuator margin (T2), and behavior-successor mismatch (T3), together with a minimal velocity-certified action set enforced in the QP. A separate corollary states the additional torque-drift and averaged-request conditions needed by the two-rate implementation. These are conditions an implementation can evaluate online.
- 4) A reproducible 111-run reduced-order study, including a matched horizon trajectory-reference-governor comparison, deterministic held-out plant perturbations, repeated anchor cases across matrices, five controller interfaces, three realization maps, eight stress scenarios, the T1–T3 audit, and explicit negative cases (parameters and complete artifacts in the supplementary archive).

The experiments establish internal operation and failure detection for this interface. The implemented comparison is an ERG-style finite-horizon trajectory-reference governor matched in model and computation budget to the

TABLE I
NOTATION

Symbol	Meaning in words	Units
τ^0	torque the nominal controller asks for	Nm
τ^{pre}	torque that would be sent <i>before</i> any clipping	Nm
τ^{app}	torque actually applied, after clipping	Nm
$\hat{\tau}$	the manager's <i>prediction</i> of τ^{pre}	Nm
$\tau_{\min}, \tau_{\max}$	actuator limits	Nm
τ_{base}	torque already spoken for: gravity, Coriolis, orientation, null-space	Nm
F_h	measured interaction force	N
$G_F(x)$	converts F_h into task-acceleration units; general form $\Lambda^{-1}(x)$, this benchmark uses $\frac{1}{m}I$	1/kg
a	requested task acceleration (the decision variable)	m/s ²
a^0	acceleration the nominal controller asks for	m/s ²
$\Delta a = a - a^0$	the correction the manager applies	m/s ²
p, v	task position and velocity, propagated directly	m, m/s
$z = [p; v]$	task state	—
$e_c = p - p_r(t)$	a controller's own tracking error against its reference $p_r(t)$; internal to the controller, not part of z	m
$\mathcal{X}$	running workspace-and-speed box on z	—
$H(x)$	maps a requested acceleration to the torque it costs	Nm s ² /m
$\mathcal{A}(x)$	accelerations that are interface-realizable legal through $H(x)$ and $\tau_{\text{base}}(x)$ — at this configuration	m/s ²
$\mathcal{A}^{\text{tight}}(x)$	same set, shrunk by the uncertainty margin	m/s ²
$\bar{\delta}_\tau$	how far the real torque can differ from the prediction	Nm
α^+	remaining acceleration authority in a given direction	m/s ²
$\mathcal{S}_v = \{y : \ y\ _\infty \leq v_{\max}\}$	the certified velocity region	—
ϵ_v	certificate margin: the one-step velocity error the certificate tolerates	m/s
$\mathcal{K}_v(y)$	the certified action set: requests from y that stay inside $\mathcal{S}_v$ after up to ϵ_v of error	m/s ²

Lyapunov/dynamic-safety-margin construction in [17]; it fixes one representative baseline, and other reference/action governors or predictive safety filters could implement the same behavior coordinates.

Notation. Everything in the paper uses the following symbols. All vector inequalities are componentwise (joint by joint).

Two set operations appear in Section VI only, and both have a plain reading: $\mathcal{X} \oplus \mathcal{Y}$ means “any point of $\mathcal{X}$ plus any point of $\mathcal{Y}$ ” (worst case over both), and $\mathcal{X} \subseteq \mathcal{Y}$ means “ $\mathcal{X}$ fits inside $\mathcal{Y}$.”

II. HOW THE METHOD WORKS

This section walks through the loop once, before any of it is formalized.

Fast loop, at the controller's own high servo rate T_f , regardless of which controller is in use. Evaluate the nominal controller to get its requested acceleration a^0 , converted to torque τ^0 through the realization map (Section IV). Add the

latest correction published by the manager, converted to torque at the *current* configuration:

$$\tau^{\text{pre}} = \underbrace{\tau^0}_{\text{controller}} + \underbrace{H(x) \Delta a}_{\text{manager's correction}},$$

then apply the final high-rate projection, the last-resort guard,

$$\tau^{\text{app}} = \text{clip}(\tau^{\text{pre}}, \tau_{\min}, \tau_{\max}),$$

which is what is sent to the robot.

Slow loop, every manager period Δt , much longer than T_f . The manager publishes a correction in *acceleration*, not a cached torque; $H(x)$ is recomputed on every fast-loop tick, so configuration drift between manager updates does not turn a stale plan into an artificial disturbance. Each update:

- 1) Roll the nominal controller forward $N = 12$ steps to get the requested accelerations $a_0^0, \dots, a_{N-1}^0$ and the predicted states.
- 2) At each predicted state, build the set of accelerations the robot can actually realize, shrunk by the uncertainty margin.
- 3) Solve one small QP for the acceleration sequence closest to the nominal request that stays inside those sets, inside the workspace box, and inside the velocity-certificate set of Section VI.
- 4) Publish the first-step correction Δa ; log the margin and directional authority.

If the nominal rollout is already feasible everywhere, the QP returns the nominal request and the correction is zero — the manager is inactive during ordinary operation.

Where the anticipation lives. There is no separate “saturation detector.” Along the nominal rollout the predicted torque is $\hat{\tau} = \tau_{\text{base}} + H(a^0 - G_F \hat{F}_h)$; when that torque enters the tightened boundary layer at any step of the horizon, the corresponding constraint becomes active and the QP is forced to move. The tightening is what makes the constraint bite *early*; the forward propagation of position and velocity is what turns a future activation into a present correction.

III. RELATED WORK AND POSITIONING

Impedance control specifies a desired motion–force relation [1], and operational-space control maps task accelerations or wrenches to joint torque [2]. Both rely on adequate actuator authority. Anti-windup methods treat saturation-induced degradation inside a particular closed loop [6], [7], while model-reference adaptive impedance control seeks to reproduce a target impedance despite uncertain dynamics [11]. Interaction-control MPC methods instead adapt impedance parameters or trajectories using learned interaction dynamics, Gaussian processes, or environment estimates [12]–[15]. The present work does not replace these controllers: it requires an upstream controller to expose a task-acceleration request and supervises whether a robot can realize it.

Reference and command governors are established add-on mechanisms that modify a reference to keep a pre-stabilized system within constraints [3]. Ambrosino *et al.* directly addressed robotic actuator saturation with a trajectory-based explicit reference governor, retaining an internal PD controller

and validating position, velocity, and saturation handling on a seven-degree-of-freedom KUKA arm [17]. Generalized action governors move the supervisor after the controller, modify an action rather than a reference, and provide all-time constraint results through safe and returnable sets under bounded uncertainty [18]. Predictive safety filters similarly accept a nominal action, use a predictive model and uncertainty description, and minimally replace unsafe actions; their terminal safety policy supports a recursive guarantee absent here [5], [16]. Consequently, retaining a controller, predicting constraints, and minimizing action modification are prior art. Our design choice is the *action representation*: heterogeneous controllers expose task acceleration, while a separate realization object maps it into configuration-dependent robot torque and exposes T1–T3 audit quantities.

Robot capability and acceleration polytopes already map joint torque boxes into task-space force or acceleration sets, and ray-to-boundary calculations already quantify directional capacity [20]. We use those ideas to construct $\mathcal{A}(x)$ and α^+ , not as new polytope theory, but as the robot-specific half of the behavior–realization interface.

The certificate argument likewise builds on approximate simulation, refinement, and contract-based design [8]–[10]. Recent transferred-control-barrier-function work uses simulation functions and explicit margins to transfer a certificate from a double integrator to full nonlinear quadrotor dynamics and enforce it with a QP filter [19]. That framework is more general and provides a substantive continuous-time transfer construction. Section VI instead gives a deliberately elementary discrete one-step specialization for actuator realization: T1 and T2 isolate whether a predicted behavior request survives the torque interface without clipping, and T3 isolates whether its physical successor stays within the behavior certificate’s error budget.

Contact-oriented governors further show the breadth of modular constraint-management architectures. The compliant ERG of Gautam *et al.* inserts a reference manager between planner and controller and uses an energy certificate for contact-friendly manipulation [21]. It differs in supervised variable and certificate, but reinforces why the present claim must be about a specific interface and audit decomposition rather than the existence of an add-on safety layer.

IV. WHAT “REALIZABLE” MEANS FOR A GIVEN ROBOT

For a torque-controlled robot,

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau + J(q)^\top F_h,$$

with $M(q) \succ 0$, h the modeled gravity/Coriolis/friction terms, and F_h the measured interaction force (2-D in this benchmark). Actuator limits are the box $\tau_{\min} \leq \tau \leq \tau_{\max}$.

The architecture asks only three things of the nominal controller: its **current requested task acceleration** a^0 ; an optional **preview**, i.e. the ability to be evaluated along a predicted trajectory; and a **bound on how wrong that preview can be**. An analytic PD or impedance law can be evaluated along predicted states directly. A learned policy can be queried on predicted observations. If no validated preview exists, zero-order hold or a learned predictor may be used, but the resulting

TABLE II
POSITIONING AGAINST THE CLOSEST METHOD FAMILIES

Approach	Quantity supervised	Main guarantee or evidence	Distinction from this work
Saturation-aware trajectory ERG [17]	Joint reference	Predicted joint/velocity constraints and KUKA validation	Uses a PD-specific reference channel; no common behavior-acceleration/realization audit
Predictive safety filter [16]	Plant input/action	Uncertainty-aware predictive filtering with terminal safety policy	More general and recursively grounded; does not isolate the acceleration-to-torque T1–T3 interface
Generalized action governor [18]	Post-controller action	State/input constraints with safe-returnable-set guarantees	General action supervision; no robot task-acceleration realization object
Task-space capacity polytope [20]	Capability set, not a supervisor	Real-time force/velocity/acceleration capacity computation	Supplies geometry used here, but not the two-rate correction or interface contract
Transferred CBF [19]	Concrete-system safety-filter input	Simulation-function-based certificate transfer to nonlinear dynamics	General continuous-time transfer theory; not specialized to clipping and torque realizability
Compliant ERG [21]	Motion reference	Contact-energy certificate	Reference and energy interface rather than behavior acceleration and T1–T3
This work	Task-acceleration request	Finite-horizon QP plus sampled one-step realization audit	Explicit behavior request/robot realization split; no recursive or hardware guarantee

Table I. Positioning against the closest method families. The comparison is architectural and theorem-level. Section VII additionally implements an ERG-style horizon trajectory-reference governor matched to this benchmark; it is not an exact reproduction of [17].

error must be included in the bound. A black-box controller with neither query access nor a bounded preview error lies outside the certificate. The induced torque τ^0 is obtained from a^0 through the realization map below.

With task state $z = [p; v]$ — absolute task position and velocity, propagated directly rather than relative to a possibly time-varying reference — the requested behavior over a local operating region is

$$\ddot{p} = a + d_{\text{res}},$$

in words: the acceleration the task actually realizes equals the requested acceleration plus a residual disturbance d_{res} . Once the interaction-force compensation of (1) below is introduced, d_{res} excludes F_h 's own (compensated) contribution and carries only what that compensation misses — residual force-model error, cross-coupling, discretization — checked explicitly as a one-step model error in Section VI. A reference-tracking controller may separately form its own tracking error $e_c = p - p_r(t)$ against a reference $p_r(t)$ to compute its request; this controller-internal error is not itself part of z .

Over one manager period this integrates to the constant-acceleration update

$$p^+ = p + \Delta t v + \frac{1}{2}(\Delta t)^2(a + d_{\text{res}}), \quad v^+ = v + \Delta t(a + d_{\text{res}}),$$

which is the standard $z^+ = Az + B(a + d_{\text{res}})$ double integrator, used here as the shared interface coordinate.

Realizing a requested acceleration a costs torque. For the full-rigid-body model, let $N_\tau(q)\tau_{\text{sec}}$ denote a dynamically consistent secondary torque satisfying $JM^{-1}N_\tau = 0$. The bias and secondary contribution required by the operational-space realization is then

$$\begin{aligned} \tau_{\text{base}}(x) &= J^\top \Lambda (JM^{-1}h - \dot{J}\dot{q}) + N_\tau \tau_{\text{sec}}, \\ H(x) &= J(q)^\top \Lambda(q), \quad \Lambda = (JM^{-1}J^\top)^{-1}, \\ G_F(x) &= \Lambda^{-1}(x). \end{aligned} \quad (1)$$

The commanded torque is

$$\tau = \tau_{\text{base}}(x) + H(x)(a - G_F(x)F_h). \quad (2)$$

Substitution into the robot dynamics gives $\ddot{p} = a$: the $JM^{-1}h$ and $\dot{J}\dot{q}$ terms cancel, the dynamically consistent secondary torque does not enter task acceleration, and $HG_F F_h = J^\top F_h$ cancels the modeled interaction-force term. Thus a is the desired *total* task acceleration, not a residual defined against a separate disturbance term. This remains conditional on the stated bias expression and dynamic consistency. Any omitted friction, imperfect bias term, non-decoupled secondary torque, or force-model error belongs in d_{res} , $\bar{\delta}_\tau$, or the one-step mismatch bound rather than being silently absorbed by the identity. The reduced-order benchmark of Section VII instantiates τ_{base} and H directly and uses the scalar approximation $G_F = \frac{1}{m}I$ (effective task-space mass m , Appendix A), exact when $\Lambda \approx mI$.

We assume $J(q)$ has full row rank throughout the verified operating region and that $JM^{-1}J^\top$ is uniformly nonsingular there, $\lambda_{\min}(JM^{-1}J^\top) \geq \underline{\lambda} > 0$; configurations violating this condition lie outside the certificate. A damped operational-space inverse may be used in implementation near such configurations, but its acceleration-realization defect must then be folded into $\bar{\delta}_\tau$ (torque domain) or the one-step model-error bound (velocity domain) rather than assumed away.

In words: $H(x)$ is the price, in joint torque, of one unit of net task acceleration at this configuration, after the force compensation of (1). A request is realizable only if that price fits in the remaining budget:

$$\tau_{\min} \leq \tau_{\text{base}}(x) + H(x)(a - G_F(x)F_h) \leq \tau_{\max}. \quad (3)$$

Call the set of a satisfying (3) the **feasible request set** $\mathcal{A}(x)$. This is the object that cannot be made robot-independent: it moves with the configuration through both $H(x)$ and $\tau_{\text{base}}(x)$, and removing that dependence would remove exactly the geometry that causes saturation.

Interface consistency. The nominal interface supplies a requested task acceleration a^0 , either directly — as every controller evaluated in Section VII does — or, for an opaque torque-only controller, through a robot-specific conversion satisfying the residual bound below. The induced nominal torque is

$$\tau^0 = \tau_{\text{base}}(x) + H(x)(a^0 - G_F(x)F_h) + r_\tau(x),$$

$$|r_\tau(x)| \leq \bar{r}_\tau,$$

where r_τ is an interface-consistency residual, bounded by the tightening margin $\bar{\delta}_\tau$ below; the five interfaces evaluated in Section VII are constructed so $r_\tau \equiv 0$.

Condition (3) is a set of parallel slabs in a , one pair per joint, so $\mathcal{A}(x)$ is already a polytope in acceleration space.

Two practical consequences follow, and only these are used later.

- **It is direction-dependent.** Authority is a polytope, not a ball: the robot can have plenty of torque left overall and still be nearly unable to accelerate along one particular direction.
- **It is cheap to probe.** Directional authority is evaluated directly from the half-space representation of $\mathcal{A}^{\text{tight}}(x)$, as defined in Section V-B; no vertex enumeration is required.

The manager predicts $\hat{\tau}$; the torque that would actually be sent, τ^{pre} , differs from it. Suppose that difference is bounded joint by joint:

$$|\tau^{\text{pre}} - \hat{\tau}| \leq \bar{\delta}_\tau. \quad (4)$$

Then it is not enough for the *prediction* to sit inside the box; the prediction plus its error must. So the manager enforces the **tightened** condition

$$\tau_{\min} + \bar{\delta}_\tau \leq \hat{\tau} \leq \tau_{\max} - \bar{\delta}_\tau, \quad (5)$$

and we write $\mathcal{A}^{\text{tight}}(x)$ for the accelerations satisfying it. Equation (5) is the whole tightening mechanism: a boundary layer of width $\bar{\delta}_\tau$ inside each actuator limit. It is what makes the constraint act *before* the physical limit rather than at it, and Section VII-E shows what happens when it is removed.

The bound $\bar{\delta}_\tau$ must cover state-estimation error, interpolation between manager updates, secondary torque, torque-rate limiting (if present — this implementation has none), and any other implementation effect that can move the pre-clip torque. The implementation uses two concrete instantiations of it: a conservative, action-independent $\bar{\delta}_\tau^{\text{QP}}$, evaluated at the acceleration limit rather than the not-yet-chosen request, inside the QP's own tightening; and a smaller, action-dependent $\bar{\delta}_\tau(a, F_h)$, evaluated at the request actually solved for, used in the (T1) audit of Section VII. Since the QP's bound is evaluated at a worst-case acceleration, $\bar{\delta}_\tau^{\text{QP}} \geq \bar{\delta}_\tau(a, F_h)$ over the acceleration box.

Problem statement. Given a nominal controller supplying a^0 , find a correction Δa such that:

- 1) $a = a^0 + \Delta a$ stays close to the nominal request;
- 2) the realized torque stays inside the actuator box over the horizon, allowing for uncertainty;
- 3) the predicted state satisfies running workspace and speed constraints; and
- 4) the result can be tested against sufficient conditions for transferring an independently established behavior certificate.

V. THE PREDICTIVE CORRECTION AND WHAT THE MANAGER REPORTS

A. The predictive correction

The nominal controller and the map $H(x)$ are evaluated, and the final torque-box projection applied, every fast-loop period T_f ; the manager — including the rate constraint on successive planned steps introduced below, which is evaluated only when the QP is solved — runs every manager period Δt , much longer than T_f . Within one manager update we write a_i for the request at horizon step $i = 0, \dots, N-1$ and a_i^0 for what the nominal controller would ask there; the manager-update index is suppressed since everything below lives inside a single update. At $i = 0$, a_{-1} denotes the manager's own solved a_0 from the *start* of the previous update, not that update's uncorrected nominal, so the rate constraint and smoothing term below bound change relative to the manager's own last decision.

The decision variable is the *complete* request a_i , not the correction — the constraints are physical conditions on the actual request, and the correction is read off afterwards. Subject to, for each i : a prediction row, a realizable-request row, a velocity-certificate row, a workspace row, and a request-rate row:

$$\begin{aligned} \min_{a_0, \dots, a_{N-1}} \quad & \sum_{i=0}^{N-1} \left(\underbrace{\|a_i - a_i^0\|_W^2}_{\text{stay close to the controller}} + \underbrace{\|a_i - a_{i-1}\|_{W_\Delta}^2}_{\text{stay smooth}} \right) \\ p_{i+1} = & p_i + \Delta t v_i + \frac{1}{2}(\Delta t)^2 a_i, \quad v_{i+1} = v_i + \Delta t a_i, \\ \tau_{\min} + \bar{\delta}_{\tau,i}^{\text{QP}} \leq & \tau_{\text{base}}(\hat{x}_i) + H(\hat{x}_i)(a_i - G_F(\hat{x}_i)\hat{F}_{h,i}) \\ & \leq \tau_{\max} - \bar{\delta}_{\tau,i}^{\text{QP}}, \\ |v_i + \Delta t a_i| \leq & v_{\max} - \epsilon_v, \quad z_{i+1} \in \mathcal{X}, \\ \|a_i\|_\infty \leq & a_{\max}, \quad \|a_i - a_{i-1}\|_\infty \leq \dot{a}_{\max} \Delta t. \end{aligned} \quad (6)$$

Every constraint is linear in a_i and the cost is convex quadratic, so this is a small dense QP. Here $\hat{x}_i$ is the state on the fixed nominal rollout used to assemble it, $\hat{F}_{h,i}$ is the forecast interaction force at horizon step i — by default the measured force held constant across the horizon (a zero-order hold; Section VII-B's preview-mismatch case is exactly the failure mode of this choice), with other preview policies evaluated in the ablations of Section VII-E — and $\mathcal{X} = \{z : |p| \leq \text{pos}_{\max}, |v| \leq v_{\max}\}$ is the running workspace-and-speed box on the task state $z = [p; v]$. The velocity-certificate row is the set $\mathcal{K}_v$ of Section VI, written out: it replaces the ordinary predicted-speed bound $v_{\max}$ inside $\mathcal{X}$ with the tighter $v_{\max} - \epsilon_v$ for one manager step ahead. Together with the realizable-set row, a feasible solution satisfies $a_i \in \mathcal{A}^{\text{tight}}(\hat{x}_i) \cap \mathcal{K}_v(v_i)$ by construction — the certified-action-set and tightened-actuator parts of Theorem 1's premise, not a check applied afterward. State, acceleration, certificate, and rate bounds are all hard; there are no slack variables and no separately constructed terminal invariant set.

The published correction is the first-step gap:

$$\Delta a = a_0 - a_0^0.$$

Interpretation. The manager first checks the nominal rollout against every constraint above; if it already satisfies all of them, the rollout is returned unchanged and $\Delta a = 0$.

Otherwise the QP is solved, returning the minimum-cost adjustment — jointly penalizing deviation from the nominal behavior and variation of the corrected request over the whole horizon — that pulls the request back onto the boundary of the feasible set.

The QP is assembled *along the frozen nominal rollout*, which keeps each realizable-set constraint linear; if that rollout is wrong — from preview mismatch, or because the QP’s own correction moves the trajectory away from the nominal one — the future geometry used by the current plan may become inaccurate. The present benchmark does not provide an analytical bound for this horizon-geometry drift — $\bar{\delta}_\tau$ and condition (T1) instead bound realization-map and model uncertainty evaluated at the current state (Section VII-D), a related but distinct error source. Frequent replanning limits how long any one horizon’s geometry is trusted, while the model- and preview-mismatch cases expose horizon-geometry drift’s consequences empirically (Section VII-B).

If the solver reports infeasibility, the implementation instead projects the first nominal acceleration onto the current tightened torque and state halfspaces (a continuous-time control-barrier-function set, not the QP’s own discrete constraint) and tiles that reactive command over the stored sequence, returning zero acceleration if even that is empty; the final high-rate projection remains active either way. This fallback gives a deterministic bounded command but does **not** recover horizon feasibility or satisfy the transfer conditions — trajectories containing fallback steps are not horizon-MPC behavior.

B. What the manager reports

For each joint, the raw torque headroom still unused; the reported margin is the worst joint:

$$\mu = \min_j \{ \tau_{\max,j} - \tau_j, \tau_j - \tau_{\min,j} \}.$$

So $\mu > 0$ means every joint is strictly inside its limits, $\mu = 0$ means some joint is exactly at a limit, and $\mu < 0$ means the request is not realizable. The benchmark uses symmetric limits, $\tau_{\min,j} = -L_j$ and $\tau_{\max,j} = L_j$, for which this reduces to $\min_j (L_j - |\tau_j|)$. μ is reported in newton-metres, not normalized by each joint’s own range — a joint with a wider torque range simply has more raw headroom to report, so μ is not comparable across robots with different actuator limits. Over the horizon we report the smallest value.

Scalar margin does not reveal whether the robot can still accelerate *in the direction it needs to*. Given a *feasible* current acceleration $a_c \in \mathcal{A}^{\text{tight}}(x)$ and a unit direction y , the remaining authority along y is how far one can move before leaving the tightened feasible set, so the indicator counts the same uncertainty margin the optimizer already enforces:

$$\alpha^+(x, a_c, y) = \max \{ \alpha \geq 0 : a_c + \alpha y \in \mathcal{A}^{\text{tight}}(x) \}.$$

$\mathcal{A}^{\text{tight}}(x)$ is a polytope defined by one tightened slab per actuator, $\{a : l_j \leq h_j(x)^\top a \leq u_j\}$, where $h_j(x)^\top$ is the j -th row of $H(x)$; it need not itself be an affine image of a box, but this is not required for α^+ : it is still a closed-form evaluation and not a search, directly from the halfspace form

$\{v : Av \leq b\}$ of $\mathcal{A}^{\text{tight}}(x)$, which any polytope in this representation admits: $\alpha^+ = \min_{j:(Ay)_j > 0} (b_j - (Aa_c)_j) / (Ay)_j$, the point where the ray from a_c along y first leaves the set. If the current request is already outside $\mathcal{A}^{\text{tight}}(x)$, the manager reports $\alpha^+ = 0$ by convention rather than apply this formula, which is meaningful only from a feasible starting point. A small α^+ is **directional authority collapse**: the robot may retain plenty of unused torque overall and still be unable to resist a push along y . This is the failure a single utilization number hides.

The manager also reports whether its horizon problem is feasible. This is a useful recoverability indicator, but it is **not** membership in a certified viability kernel, because no terminal invariant set or backup policy is constructed. Section VII therefore uses the term *near-boundary braking stress case* rather than *point of no return*.

VI. A CONDITIONAL REALIZATION CONTRACT

Sections IV–VI describe one robot. This section asks what survives when the robot changes. It specializes refinement logic to the acceleration-to-torque interface and states exactly which implementation quantities must hold for one transition.

A. The setup in words

Suppose someone has already proved something about the *behavior* — for instance, that velocity never leaves a set $\mathcal{S}_v$, provided each step’s request comes from some *acceptable* set of actions rather than one fixed policy, and the one-step model error stays within a margin ϵ_v . Call that the **certificate**. It is stated over the behavior alone, independent of any particular robot, and it ranges over an acceptable set of actions rather than one fixed policy: different robots, or the same robot at different moments, pick different members of that set, which is what makes the set — rather than any one policy built on top of it — the reusable object.

Now put a real robot underneath. Three things can go wrong:

- 1) the manager’s torque prediction $\hat{\tau}$ may be wrong;
- 2) the resulting torque may not fit in the actuator box, so clipping activates and the robot stops doing what was requested;
- 3) even without clipping, the real robot’s one-step velocity may differ from the double-integrator prediction by more than the certificate tolerates.

Theorem 1 records the resulting interface contract: rule out all three for whatever request the manager actually picked from the acceptable set, and the one-step certificate conclusion follows.

B. The certified action set, and why velocity only

Let $y = v$ be the task velocity and $F_v(y, a) = y + \Delta t a$ its one-step update (the velocity half of the double integrator in Section IV). Define the certified region, with the margin ϵ_v satisfying $0 < \epsilon_v < v_{\max}$,

$$\mathcal{S}_v = \{y : \|y\|_\infty \leq v_{\max}\}, \quad \mathcal{E}_v = \{d : \|d\|_\infty \leq \epsilon_v\},$$

and, at every $y \in \mathcal{S}_v$, the **certified action set**

$$\mathcal{K}_v(y) = \{a : \|F_v(y, a)\|_\infty \leq v_{\max} - \epsilon_v\}.$$

In words: $\mathcal{K}_v(y)$ is every request that keeps next step's velocity inside the certified region even after absorbing a one-step model error of size up to ϵ_v . Position is deliberately left out of the certificate — it stays a plain running constraint in $\mathcal{X}$ — because velocity is the only quantity this benchmark actually measures and audits a one-step mismatch for (Section VII-D); tightening position too would not be backed by a measured quantity.

$\mathcal{K}_v$ is never empty on $\mathcal{S}_v$ whenever $\epsilon_v/\Delta t$ is within the abstract model's admissible acceleration range: decelerating at that rate removes ϵ_v of speed within one manager period, so a legal request back into the tighter band always exists, including when several velocity components sit on the boundary at once (Appendix A gives the margin this leaves against the benchmark's own acceleration bound). This is nonemptiness of $\mathcal{K}_v(y)$ alone, in the abstract velocity model; its intersection with the robot-specific tightened torque set $\mathcal{A}^{\text{tight}}(x)$ and the rate and state constraints may still be empty, which is exactly what the reactive fallback of Section V-A exists to handle, and does happen in the experiments of Section VII.

C. The one-step interface lemma

Write $\Pi(x) = y = v$ for the map from the physical state to the task velocity the certificate is stated in, and let $f^d(x, \tau, F_h)$ denote the one-step physical dynamics: the state the robot actually reaches, one manager period later, after applying torque τ under force F_h from state x .

Theorem 1 (One-step conditional velocity-certificate realization): Suppose the physical state x lies in a verified operating region, $\Pi(x) \in \mathcal{S}_v$, and the request a satisfies $a \in \mathcal{K}_v(\Pi(x))$ — any such a , not one distinguished policy. If:

(T1) Prediction accuracy. The torque that would be sent is within the stated bound of the manager's prediction,

$$|\tau^{\text{pre}} - \hat{\tau}| \leq \bar{\delta}_\tau.$$

(T2) Actuator-margin test. The prediction, widened by that bound, still fits inside the actuator box,

$$\tau_{\min} + \bar{\delta}_\tau \leq \hat{\tau} \leq \tau_{\max} - \bar{\delta}_\tau.$$

(T3) Certificate-margin test. The one-step velocity mismatch between the real robot and the double-integrator prediction fits inside the certificate margin,

$$\|\Pi(f^d(x, \tau^{\text{pre}}, F_h)) - F_v(\Pi(x), a)\|_\infty \leq \epsilon_v.$$

then clipping does not alter the frozen torque input represented in f^d , $\tau^{\text{app}} = \tau^{\text{pre}}$, and the corresponding physical successor stays in the certified region,

$$\Pi(f^d(x, \tau^{\text{app}}, F_h)) \in \mathcal{S}_v.$$

Proof: (T1)+(T2) give $\tau_{\min} \leq \tau^{\text{pre}} \leq \tau_{\max}$, so the clip is inactive and $\tau^{\text{app}} = \tau^{\text{pre}}$. (T3) places the real one-step successor inside $F_v(\Pi(x), a) \oplus \mathcal{E}_v$. This set lies in $\mathcal{S}_v$ by the definition of $\mathcal{K}_v$. ■

Corollary 1 (Conditional two-rate interval realization): Let manager update k span $[t_k, t_{k+1}]$, and let $a_f(t)$ be the time-varying fast-loop request produced by reevaluating the nominal controller while holding Δa_k . Define its interval average

$$\bar{a}_k = \frac{1}{\Delta t} \int_{t_k}^{t_{k+1}} a_f(t) dt$$

(or the corresponding average of the twenty 1 kHz samples in the implementation). Suppose $\bar{a}_k \in \mathcal{K}_v(\Pi(x_k))$, and a componentwise inter-update bound β_k satisfies

$$|\tau^{\text{pre}}(t) - \tau^{\text{pre}}(t_k)| \leq \beta_k \quad \text{for every } t \in [t_k, t_{k+1}].$$

If (T1) holds at t_k , (T2) is strengthened to

$$\tau_{\min} + \bar{\delta}_{\tau,k} + \beta_k \leq \hat{\tau}_k \leq \tau_{\max} - \bar{\delta}_{\tau,k} - \beta_k,$$

and the endpoint defect relative to $F_v(\Pi(x_k), \bar{a}_k)$ is at most ϵ_v , then clipping is inactive at every fast-loop sample in the interval and $\Pi(x_{k+1}) \in \mathcal{S}_v$.

Proof: The triangle inequality combines (T1) with β_k , placing every inter-update $\tau^{\text{pre}}(t)$ inside the actuator box. The interval-average definition gives the double-integrator velocity update $F_v(\Pi(x_k), \bar{a}_k)$; the endpoint argument is then identical to Theorem 1. ■

Theorem 1 specializes robust one-step refinement to the actuator interface: (T1)–(T2) certify that the frozen behavior request survives the actuator interface without clipping, and (T3) supplies the abstract-to-physical successor margin. Corollary 1 gives the corresponding sufficient condition for the implemented two-rate signal; the present benchmark measures its operating-region bound β_k from the 1 kHz samples, with the per-source decomposition below as a route to an analytic β_k . Both results are one-step; a continuous-time transferred barrier or recursive safety filter, as in simulation-function-based transferred CBFs [19], is a separate construction.

D. What actually transfers

The theorem requires, for each new robot, its own Π , $\hat{\tau}$, the bound $\bar{\delta}_\tau$, the mismatch bound, and the operating region. What transfers unchanged is the certificate itself — the behavior model, the certified action set $\mathcal{K}_v$, the set $\mathcal{S}_v$, and the margin ϵ_v .

Condition (T3) can be decomposed into individual error sources, offering a route to an analytical bound complementing the aggregate empirical comparison the experiments use (Section VII):

$$\underbrace{\bar{\eta}_{\text{disc}}}_{\text{discretization}} + \underbrace{\bar{\eta}_{\text{hold}}}_{\text{ZOH}} + \underbrace{\bar{\eta}_{\text{sec}}}_{\text{secondary}} + \underbrace{L_F \bar{\delta}_F}_{\text{force-model}} + \underbrace{L_\tau \bar{\delta}_\tau}_{\text{torque}} \leq \epsilon_v,$$

where L_F and L_τ convert a force-model error and a torque error into the resulting one-step velocity error.

VII. EXPERIMENTS

A. Protocol

The benchmark is deterministic and uses a two-dimensional interaction task. The nominal controller and final projection run at 1 kHz; the manager uses a 50 Hz predictive rate and

horizon $N = 12$, with the 1 kHz final projection providing inter-update actuator protection; adequacy of the slower rate is operating-region and model dependent. Three configuration-dependent realization maps are evaluated: a planar 2R map, an FR3-inspired surrogate, and a six-axis-arm surrogate. The latter two are configuration-dependent surrogates reproducing different actuator geometries and limits. The manager’s uncertainty envelope is fixed from declared coefficient boxes before simulation; deterministic held-out plant coefficients, phases, and frequencies are then generated from separate seeds and never supplied to prediction or tightening. The injected waveform is therefore held out from the bound, giving a synthetic uncertainty-class test.

The behavior model, predictive objective, and audit threshold $\epsilon_{\text{audit}} = 0.03$ m/s are held fixed throughout, equal to the design radius ϵ_v of Section VI. The simulation constructs and enforces the velocity certificate $(\mathcal{S}_v, \mathcal{K}_v, \epsilon_v)$; the position constraint stays a plain running constraint in $\mathcal{X}$. Future actuator-limit scales are supplied exactly from each scenario’s schedule throughout the benchmark, not only in the horizon-ramp case of Fig. 1; the manager therefore anticipates a known or forecast limit schedule. Interaction-force preview is different: unless an oracle ablation is selected, the measured force is held constant over the horizon (Section V-A).

The five nominal-controller interfaces are PD, impedance, a small trained policy, a fitted neural policy, and a conditioned motion primitive (training and fitting details in Appendix A). The last two exercise the command/preview interface.

The 111 cases comprise 40 scenario cases, 30 controller-interface cases, 24 cross-realization cases, and 17 ablations grouped into eight families (full accounting in Appendix A).

Reported quantities are pre-clip torque excess, applied torque excess, workspace excess, behavior-realization RMSE, warning lead time, directional authority, observed one-step mismatch, and computation time. For a two-dimensional residual r_k , RMSE is the pooled component-wise value $\sqrt{\frac{1}{2K} \sum_k \|r_k\|_2^2}$. The mismatch in Table V is instead the maximum ℓ_∞ defect required by Theorem 1’s (T3), and is reported separately from the component-wise RMSE values. Applied torque remains inside its box whenever the final projection is active.

B. Anticipatory saturation management

Fig. 1 isolates why anticipation is a different constraint from reactive limiting:

- The manager’s horizon rows query the scenario’s scheduled actuator-budget drop directly, so Fig. 1 demonstrates the horizon-versus-one-step mechanism under a known future constraint.
- The first-step-only constraint’s predicted future torque violation reaches 3.587 Nm; the full-horizon constraint stays at zero throughout.
- The first-step-only variant’s QP is feasible for only 75% of updates, versus 100% for the full-horizon variant.
- The resulting 31.180 mm of workspace excess occurs entirely during the infeasible 25%, mediated by the

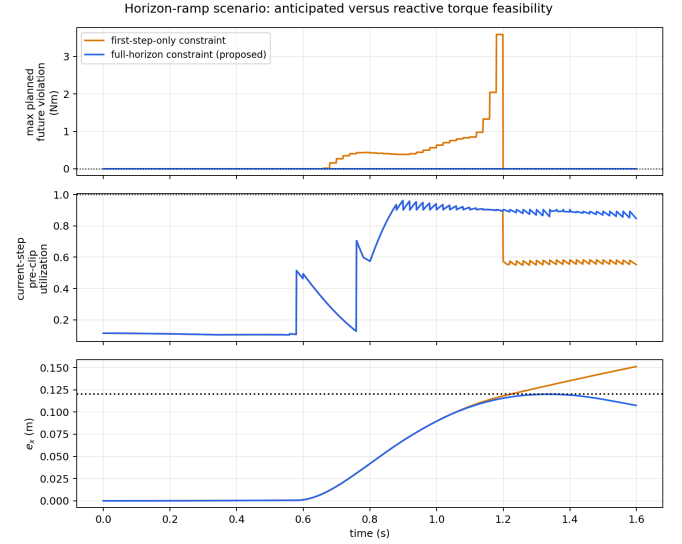


Fig. 1. Horizon-ramp scenario: a first-step-only constraint sees a legal present and walks into an illegal future; the full-horizon constraint sees it coming.

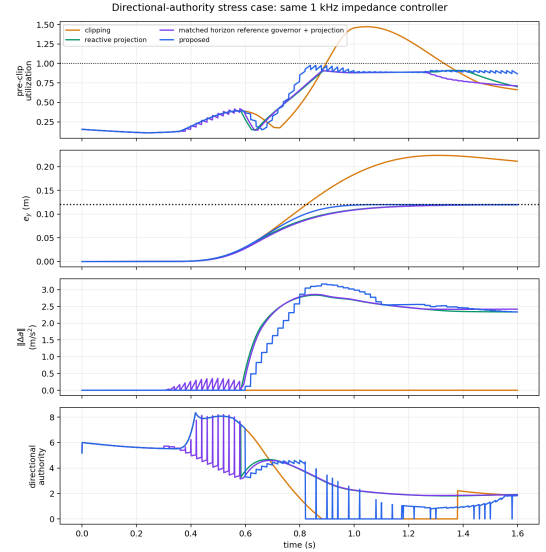


Fig. 2. Directional-authority stress case with a common impedance-controller interface.

reactive fallback of Section V-A (0.000 mm while the QP is feasible).

Fig. 2 shows the directional-authority stress case. Direct clipping exceeds both the feasible request set and the workspace bound. The predictive methods intervene earlier, and the vector correction preserves the workspace constraint while leaving a visible intervention residual. Measured along the disturbance direction (metric defined in Appendix A), the proposed manager’s authority reaches zero for about a fifth of the trajectory while the applied request keeps the workspace constraint satisfied throughout; the reactive projection and matched horizon reference governor retain positive authority in this direction across the whole run (minimum 1.82 and 1.87 m/s²).

Fig. 3 shows near-boundary braking: an outward velocity

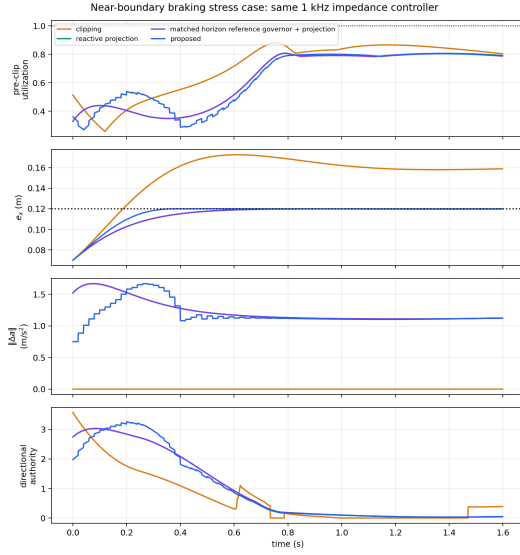


Fig. 3. Near-boundary braking stress case.

close to the position boundary under a shrinking torque budget. Direct clipping overshoots the boundary and settles outside it. The reactive projection, matched horizon trajectory-reference governor plus projection, and proposed manager all arrest the position at the boundary within the $N\Delta t = 0.24$ s horizon, using finite-horizon constraint handling without a viability kernel or terminal invariant set.

Table III reports clipping, the matched horizon trajectory-reference governor, and the proposed behavior-coordinate manager across the remaining seven scenarios. The governor uses the same realization model, 20 ms update, 0.24 s horizon, actuator-limit schedule, tightening, and 1 kHz final projection. It searches one scalar reference multiplier, whereas the proposed manager optimizes an acceleration vector at each horizon step. This makes it an ERG-style same-model numerical comparator matched to the setup of [17].

The two predictive architectures have the same qualitative outcome in the nominally successful cases: both remove pre-clip excess and hold workspace excess to numerical-scale values. Their intervention lead differs by at most 33 ms, comparable to two shared 20 ms manager updates; the horizon is $N\Delta t = 0.24$ s. Correction RMSE (governor/proposed) is 1.124/1.060, 1.389/1.401, and 0.878/0.844 m/s^2 in slow saturation, directional collapse, and near-boundary braking, respectively; in the horizon-ramp isolation case it is 0.695/0.506 m/s^2 . Thus the vector interface changes the requested behavior less in three of these four cases, while the scalar governor is marginally lower in directional collapse. This is evidence of a representation-dependent tradeoff between the vector and scalar interfaces.

Negative cases. QP feasibility is 100% in the four successful stress cases but only 92.5%, 45.0%, and 40.0% under sudden disturbance, model mismatch, and preview mismatch, with the reactive fallback of Section V-A carrying most of the resulting excess. Under preview mismatch, the correction acts on a force forecast that misses a sign change inside the horizon

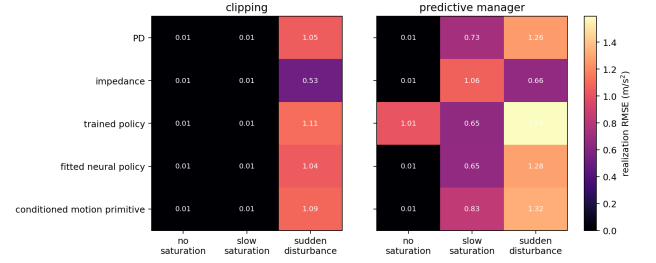


Fig. 4. Behavior-realization residuals for the five nominal-controller interfaces.

and workspace excess rises from 190.513 to 346.871 mm — substantially worse than doing nothing; model mismatch is similarly worse, 197.006 versus 190.904 mm, because the feasible geometry is assembled along a wrong rollout. Under sudden disturbance the implemented preview holds the measured force constant, so an unannounced impulse pushes pre-clip excess from 7.151 to 11.879 Nm even though the final projection keeps applied torque legal. The audit rejects all three cases, flagging exactly where Theorem 1’s premises are not met.

C. Controller-interface substitution

As shown in Fig. 4, the manager formulation and weights are unchanged across interfaces. Under no saturation, correction RMSE is below 0.01 m/s^2 for four of five controllers; the small evolution-strategy policy is the exception at ≈ 1.01 m/s^2 (bias source in Appendix A). The manager holds the state inside the workspace box but is compensating for a biased interface rather than remaining inactive. The case is kept as an interface stress test — the architecture targets realizability rather than behavior-policy quality, though its running constraints may incidentally reject a biased request.

Table IV separates the manager’s deliberate correction ($a_{\text{req}} - a^0$) from the tracking-defect RMSE ($a_{\text{actual}} - a_{\text{req}}$) most directly tied to certificate transfer. The tracking defect is 0.0111 m/s^2 for every interface — a fixed, controller-independent unmodeled-error floor, since no clipping is active under slow saturation — while correction RMSE varies with what each controller requests. All five cases pass the audit.

D. Sampled interface audit across realization maps

Each record is indexed by one feasible manager update, excluding ticks following an infeasible (fallback) solve since Theorem 1 concerns the QP-optimized plan. (T1) and (T2) are evaluated at the beginning of the interval, using the manager’s own first-horizon-step prediction $\hat{\tau}_0$ and the torque realized from that same request. Both use the same state, force, and request, though they are not numerically equal because their gap is exactly what (T1) bounds. The (T2) entry is that tick’s first-step slack, not the horizon-wide minimum reported elsewhere. (T3) uses the physical state observed at the *next* manager update. The table therefore reports a sampled start/end audit — 316 records per robot from the no-saturation

TABLE III
SAME-MODEL SCENARIO COMPARISON. C/G/P DENOTES CLIPPING/MATCHED HORIZON TRAJECTORY-REFERENCE GOVERNOR/PROPOSED BEHAVIOR-COORDINATE MANAGER.

Scenario	Pre-clip C/G/P (Nm)	Workspace C/G/P (mm)	Lead G/P (s)	P feasible	P audit
No saturation	0.000 / 0.000 / 0.000	0.000 / 0.000 / 0.000	– / –	100%	Yes
Slow saturation	0.000 / 0.000 / 0.000	78.970 / 0.000 / 0.001	0.399 / 0.414	100%	Yes
Sudden disturbance	7.151 / 10.236 / 11.879	0.000 / 0.000 / 0.000	0.086 / 0.084	92.5%	No
Directional collapse	0.708 / 0.000 / 0.000	103.868 / 0.000 / 0.003	0.307 / 0.340	100%	Yes
Near-boundary braking	0.000 / 0.000 / 0.000	52.537 / 0.000 / 0.011	0.574 / 0.568	100%	Yes
Model mismatch	12.453 / 1.190 / 2.018	190.904 / 160.711 / 197.006	0.542 / 0.633	45.0%	No
Preview mismatch	19.777 / 3.537 / 4.428	190.513 / 357.209 / 346.871	0.173 / 0.169	40.0%	No

Table II. Same-model scenario comparison. C/G/P denotes clipping/matched horizon trajectory-reference governor/proposed behavior-coordinate manager. Lead is relative to each predictive method’s own first nominal-limit event; Appendix A explains the shared-event diagnostic.

TABLE IV
CONTROLLER SUBSTITUTION UNDER SLOW SATURATION.

Interface	Correction RMSE (m/s ²)	Tracking defect RMSE (m/s ²)	Workspace excess (mm)	Audit
PD	0.726	0.0111	0.001	Pass
Impedance	1.060	0.0111	0.001	Pass
Trained policy	0.652	0.0111	0.016	Pass
Fitted neural policy	0.649	0.0111	0.001	Pass
Cond. motion primitive	0.830	0.0111	0.006	Pass

TABLE V
SAMPLED START/END QUANTITIES ASSOCIATED WITH THE THREE CONDITIONS OF THEOREM 1.

Realization map	Max. T3 defect ℓ_∞ (m/s)	Min. T1 slack (Nm)	Min. T2 slack (Nm)
Planar 2R	0.006815	0.007604	0.143756
FR3-inspired surrogate	0.006796	0.003888	0.646664
Six-axis-arm surrogate	0.006796	0.004871	0.707748

and slow-saturation cross-realization runs — over discrete manager updates; the continuous 1 kHz behavior over the intervening 20 ms is covered by the fast-loop audit described next.

The pass/fail label in Tables III and IV comes from a separate, broader unpaired audit over every 1 kHz fast-loop sample. It uses a Euclidean (T3) bound, conservative relative to Table V’s ℓ_∞ value, checks the torque-error envelope, and directly requires realized pre-clip torque to remain inside the untightened actuator box. Thus the statement that pre-clip torque remains legal throughout each passing trajectory is supported by the fast-loop check, not inferred from Table V’s manager-update snapshots alone. Table V shows positive sampled slacks across all three realization maps: the (T3) defect remains below the 0.03 m/s certificate margin, while (T1) and (T2) retain room against $\delta_\tau(a, F_h)$ (Appendix A). The true-plant coefficients are held out from the manager and generated independently of the bound waveform. The coefficient box and held-out plants are synthetic: this is a held-out uncertainty-class test, stronger than a self-envelope consistency check.

E. Ablation summary and computation

Fig. 5 isolates the four design choices central to the paper. Full-horizon enforcement keeps the horizon-ramp QP feasible

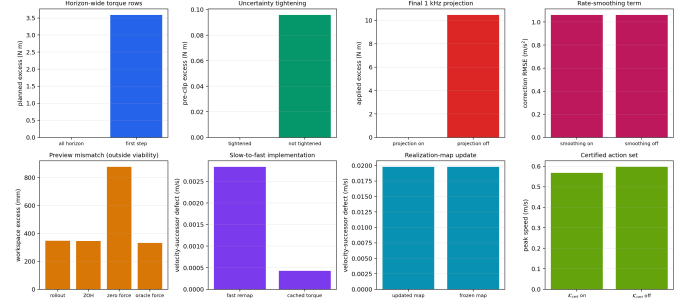


Fig. 5. Paired ablations of the predictive and fast-path implementation choices.

and removes the 3.587 Nm planned violation left by first-step-only enforcement. Removing uncertainty tightening creates 0.0958 Nm of pre-clip excess; removing the final projection creates 10.456 Nm of applied excess under sudden disturbance. In the dedicated velocity-bound case, enforcing $\mathcal{K}_v$ limits peak speed to 0.5678 m/s, versus 0.5968 m/s without it. Appendix A gives the complete 17-run ablation ledger, secondary results, and wall-clock computation figures.

VIII. DISCUSSION

Scope of the contribution. The double integrator, the QP, the ray-to-polytope authority calculation, and one-step certificate refinement are established ingredients; the contribution is their arrangement into an interface with an explicit, online-checkable boundary between behavior preservation and physical realization, together with the conditions (T1)–(T3) that decide when that boundary holds. Theorem 1 specializes robust one-step refinement to this interface, and the benchmark is a deterministic reduced-order study.

The experiments support the interface claim: the behavior-coordinate optimization and empirical audit threshold remain unchanged across the implemented controller interfaces and realization maps, while each map reconstructs its own feasible set and error bounds. This separation is the demonstrated contribution. Action governors and predictive safety filters could adopt the same coordinates; Table I places that relationship.

Reference governors, action governors, and predictive safety filters could adopt the same behavior coordinates. What this implementation contributes is an explicit software and verification boundary: a behavior request and certified action set on

one side; a robot-specific torque map, uncertainty margin, and successor audit on the other. In the present study the velocity certificate $(\mathcal{S}_v, \mathcal{K}_v, \epsilon_v)$ is deliberately minimal. The physical containment it requires — that the real one-step mismatch stays inside ϵ_v — is sampled along experiment trajectories. The torque audit separates the declared coefficient box from deterministic held-out true-plant coefficients (both synthetic). The results therefore validate a held-out perturbation inside a prescribed uncertainty class and show audit rejection under constructed mismatch.

The final high-rate projection is a distinct responsibility from behavior preservation: clipping protects the actuator, while exact realization of the requested behavior is what the manager preserves. Theorem 1 records sufficient conditions for clipping to remain inactive and for the velocity conclusion to hold for one transition. The mismatch experiments show these properties diverging. The matched horizon reference-governor comparison shows comparable constraint handling and a scenario-dependent correction tradeoff. The results establish one-step realization and a scenario-dependent comparison against a matched ERG-style governor; recursive-safety guarantees and exact reproductions of [17], a generalized action governor, or a predictive safety filter are separate evaluations.

The three realization maps test software substitution and distinct actuator geometry inside one reduced-order model class. Corollary 1 and the horizon-drift term of Section V-A state the additional bounds required to extend this to physical robot dynamics.

IX. CONCLUSION

This paper introduced a behavior-coordinate realization interface for predictive saturation management. The nominal controller exposes a task-acceleration request at its high servo rate; a slower manager forecasts robot-specific loss of realizability and modifies that request before clipping is required. The contribution is the concrete separation between a reusable behavior-side request and action set and robot-specific objects that must be reconstructed and audited: the acceleration-to-torque map, actuator margin, uncertainty bounds, and operating region. A minimal velocity-certified action set is enforced inside the QP as an existence witness, while Theorem 1 states the corresponding one-step realization contract.

Across 111 deterministic reduced-order runs, full-horizon enforcement eliminates a planned future violation that remains in the first-step-only variant, and the same interface accepts five nominal-controller forms. A sampled audit applies the same behavior-mismatch threshold while retaining distinct realization maps for three robot surrogates. Slow saturation, directional authority collapse, and near-boundary braking are handled within the tested region. A matched ERG-style trajectory-reference governor handles those constraints comparably; the vector behavior correction has lower correction RMSE in three of the four successful or horizon-isolation cases, but not in directional collapse. Abrupt disturbance and severe mismatch instead make the contract fail and frequently render the QP infeasible, leaving only the reactive fallback

and final projection. These results establish that the interface operates, exposes a measurable representation tradeoff, and rejects assumptions as designed.

Exact reproductions of saturation-aware ERG, generalized action-governor, or predictive-safety-filter implementations, and validation with independently identified error bounds on full rigid-body realizations, are the next evaluations; position certificates and recursive feasibility are separate theoretical directions built on the same behavior/realization boundary.

APPENDIX A BENCHMARK PARAMETERS

This appendix lists the constants needed to reproduce the numbers in Section VII, including the complete ablation ledger; the full realization-map functions $H(x)$, $\tau_{\text{base}}(x)$, the learned-policy weights, and the per-scenario timelines are in the simulation code released alongside this paper.

Shared constants (BenchmarkConfig, simulation/saturation_benchmark.py): fast period $T_f = 1$ ms, manager period $\Delta t = 20$ ms, horizon $N = 12$, episode duration 1.6 s, workspace box $\text{pos}_{\text{max}} = 0.12$ m, speed box $v_{\text{max}} = 0.60$ m/s, acceleration bound $a_{\text{max}} = 12.0$ m/s², acceleration rate bound $\dot{a}_{\text{max}} = 120.0$ m/s²/s, certificate/audit margin $\epsilon_v = \epsilon_{\text{audit}} = 0.03$ m/s, cost weights $W = 1.0 \cdot I$, $W_{\Delta} = 0.025 \cdot I$, OSQP tolerance 10^{-5} . These give $\epsilon_v/\Delta t = 1.5$ m/s², well under a_{max} — the concrete margin behind $\mathcal{K}_v$'s nonemptiness (Section VI).

Realization maps (torque limits in Nm, mass in kg): every listed torque-limit array is the positive magnitude L of a symmetric box, $\tau_{\text{min}} = -L$ and $\tau_{\text{max}} = L$. Planar 2R — 2 joints, $L = [9.0, 7.5]$, mass 2.2; FR3-inspired surrogate — 7 joints, $L = [16.0, 13.0, 12.0, 15.0, 6.0, 6.0, 4.5]$, mass 3.0; six-axis-arm surrogate — 6 joints, $L = [12.0, 10.0, 9.0, 8.5, 5.5, 4.5]$, mass 2.6. The interaction-error bound $\bar{\delta}_{\tau}$ uses base term 0.03 Nm per joint and the H -dependent term below, with $H_0 = H(x)$ evaluated once at each map's zero-state configuration.

Cross-realization audit detail. Before the simulated true plants are instantiated, the manager's per-joint bound is fixed as

$$\bar{\delta}_{\tau}^{(j)} = s_{\text{mm}} \left(\underbrace{0.03}_{\text{Nm, base}} + 0.008 \sum_k \max(|H_0|_{jk}, 0.25) \left| a_k - \frac{F_{h,k}}{m} \right| \right),$$

where s_{mm} is the scenario's mismatch-scale multiplier (1.0 for the cross-realization cases reported here, up to 2.4 for model mismatch elsewhere in Section VII), ranging from 0.0300 to 0.0926 Nm over the audit; this is $\bar{\delta}_{\tau}(a, F_h)$, the action-dependent bound of Section IV, not the QP's own more conservative $\bar{\delta}_{\tau}^{\text{QP}}$. The held-out true plants are then generated with seeds 31, 37, and 41 for planar 2R, FR3-inspired, and six-axis-arm maps. Their independent coefficients satisfy $|c_H| \leq 0.0065 < 0.008$ and base-error amplitudes in $[0.012, 0.027]$ Nm < 0.03 Nm, with independently sampled

phases and frequencies; these realized values are never passed to the manager. Table V reports the resulting (T1) and (T2) slacks directly. This is a deterministic held-out test within a prescribed synthetic coefficient box.

Baseline architectures. Reactive 1 kHz projection: exact Euclidean projection onto the intersection of the current-step tightened torque halfspaces and a relative-degree-two box control-barrier-function halfspace set with decay rate $\lambda = 8$ (`reactive_state_halfspaces`), i.e. position rows $\mp a \leq \pm 2\lambda v + \lambda^2(\text{pos}_{\max} \mp p)$ and speed rows $\mp a \leq \lambda(v_{\max} \mp v)$. Matched horizon trajectory-reference governor: at each manager tick, `governor_scale` uses bisection to find the largest $\alpha \in [0, 1]$. For each candidate it rolls the state through all $N = 12$ steps, evaluates the nominal controller against $p + \alpha(p_r - p)$, and checks the same scheduled actuator limits, uncertainty-tightened torque halfspaces, workspace bound, measured-force hold, model, 20 ms update, and 0.24 s horizon as the proposed manager. One bound is intentionally left in the comparator’s favor: the governor is checked against the untightened speed limit $v_{\max} = 0.60$ m/s, whereas the proposed manager additionally enforces the velocity certificate and is therefore held to $v_{\max} - \epsilon_v = 0.57$ m/s, a 5% tighter bound. The comparison is thus conservative toward the proposed method: its lower correction RMSE in Table III is obtained under a strictly smaller admissible set than the comparator enjoys. The first governed reference is held until the next manager tick and its requested acceleration is passed through the same reactive projection. This is an ERG-style same-model comparator motivated by [17]; it directly searches finite-horizon constraint feasibility.

Directional-authority metric. α^+ (Section V-B) is a function of a chosen direction y . Fig. 2 reports it along the fixed, exogenous disturbance direction, which is comparable across methods and time; the manager’s own correction direction is endogenous and can change discontinuously as its output changes. Both use the same halfspace-ray construction.

Governor-search validation. The matched governor uses bisection over α , which assumes feasibility is monotone in α . A dense 201-point grid check across all 160 manager ticks at which the governor is invoked in this benchmark found no non-monotone case. A regression test additionally confirms that its horizon attenuates the reference before a scheduled limit drop that a one-step version cannot yet see.

Controller interfaces. PD and impedance are closed-form analytic laws. The trained policy is a small tanh-squashed single-hidden-layer network fit by a deterministic evolution strategy. The fitted neural policy is an 18-unit tanh hidden layer with a linear output head, least-squares fit to impedance-law demonstrations (`saturation_benchmark.py`, `RLPolicyController/neural-controller` fit). The conditioned motion primitive is a PD servo tracking a primitive-conditioned reference.

Trained-policy bias (Section VII). The evolution-strategy policy’s limited training set does not cover the test trajectory symmetrically, and it retains a positive- y command bias; this is an observed generalization error of that particular fit.

Computation. In the saved run, both the manager’s and the fast path’s worst observed wall-clock maxima remained

below their nominal periods (20 ms and 1 ms respectively), with median solve times well under either budget. These are wall-clock measurements of computational cost on a development machine, not a real-time guarantee; because they are nondeterministic run-to-run, the released `results/all_experiment_metrics.json` is the source of record rather than any specific figure quoted here. The simulator itself advances the physical state by exactly 1 ms every step and does not model a scheduler that drops or delays steps on a missed deadline. A hard real-time implementation with an explicit scheduling policy is future work.

Case accounting. The 111 cases comprise 40 scenario cases, 30 controller-interface cases, 24 cross-realization cases, and 17 ablations grouped into eight families. The 40 scenario cases are eight scenarios evaluated with five channels: four architectures — direct clipping, a reactive 1 kHz projection, the matched horizon trajectory-reference governor followed by the same reactive projection, and the proposed horizon-wide behavior-coordinate correction with final projection — plus one unconstrained nominal-reference channel without torque projection. The stress scenarios are no saturation, slow saturation, sudden disturbance, directional authority collapse, near-boundary braking, model mismatch, preview mismatch, and a horizon-ramp scenario reported directly as Fig. 1 and used for the constraint-width ablation of Section VII-E. A ninth scenario, starting inside $\mathcal{S}_v$ near its tightened boundary, is used only for the velocity-certificate ablation and enters neither the scenario nor the cross-realization matrices.

Complete ablation ledger. All ablations use the FR3-inspired surrogate and impedance controller.

Secondary ablation results. Removing request smoothing leaves correction RMSE under slow saturation essentially unchanged (1.0603 m/s² without it versus 1.0604 m/s² with it). Zero-force preview increases workspace excess from 346.871 to 875.475 mm; in the severe mismatch case, no preview option restored viability. Recomputing the fast map matched cached torque in this reduced-order benchmark, with updating and freezing numerically indistinguishable.

Reproducibility. All code, tests, saved metrics, and figure-generation scripts are released alongside this paper. The complete configuration — including the ablation-family definitions and scenario force/goal timelines — is fixed in `simulation/saturation_benchmark.py` and `simulation/run_all_experiments.py`. The evolution-strategy policy uses RNG seed 4, and the fitted neural policy uses seed 7. From the `pHRI/saturation` directory, the complete deterministic suite is regenerated with:

```
python3 -m pip install -r requirements.txt
MPLCONFIGDIR=/tmp/mpl-saturation \
XDG_CACHE_HOME=/tmp/cache-saturation \
PYTHONPATH=simulation \
python3 simulation/run_all_experiments.py
```

The command writes `results/all_experiment_metrics.json` and the figures used in Section VII.

APPENDIX B

AN A PRIORI BOUND FOR CONDITION (T3)

Section VI verifies (T3) by measurement. This appendix discharges the decomposition sketched there into an *a priori*

TABLE VI
COMPLETE ABLATION LEDGER

#	Scenario	Variant	Changed element
1	Horizon ramp	Full	Complete proposed manager
2	Horizon ramp	First-step torque	Torque constraints only at the first horizon step
3	Slow saturation	Cached torque	Hold the complete torque command between manager updates
4	Sudden disturbance	Full	Complete proposed manager
5	Sudden disturbance	No final projection	Disable the 1 kHz torque projection
6	Model mismatch	Full	Complete proposed manager
7	Directional collapse	Full	Complete proposed manager
8	Directional collapse	No tightening	Set $\bar{\delta}_\tau = 0$ in the predictive constraints
9	Model mismatch	Frozen realization map	Freeze H and τ_{base} at the current manager state
10	Model mismatch	Updated realization map	Evaluate the maps along the nominal predicted states
11	Preview mismatch	Full	Measured-force hold preview
12	Preview mismatch	Acceleration ZOH	Hold the current nominal acceleration over the horizon
13	Preview mismatch	Zero-force preview	Set the forecast force to zero
14	Preview mismatch	Oracle-force preview	Supply the scenario's future force
15	Slow saturation	No smoothing	Set $W_\Delta = 0$
16	Certificate-margin case	Constrained	Enforce $\mathcal{K}_v$
17	Certificate-margin case	Unconstrained	Remove $\mathcal{K}_v$

sufficient condition: if the five constants below are known for a given robot and operating region, (T3) holds without being checked online. The benchmark of Section VII measures the aggregate defect; this appendix gives an *a priori* route that could replace that measurement.

A. Assumptions

Throughout, $\mathcal{O}$ denotes the verified operating region, all norms are ℓ_∞ on task quantities and induced norms on matrices, and Δt is the manager period with T_f the fast period ($\Delta t = n_f T_f$).

(A1) **Regularity.** $J(q)$ has full row rank on $\mathcal{O}$ with $\lambda_{\min}(JM^{-1}J^\top) \geq \underline{\lambda} > 0$ and $\lambda_{\max}(JM^{-1}J^\top) \leq \bar{\lambda}$; also $\sigma_{\max}(J) \leq \bar{\sigma}_J$ and $\lambda_{\min}(M) \geq \underline{\mu} > 0$. This is the standing assumption of Section IV, quantified.

(A2) **Bounded motion.** $\|\dot{x}\| \leq V_x$ on $\mathcal{O}$.

(A3) **Lipschitz realization.** The map from state to realized task acceleration under a held torque is Lipschitz on $\mathcal{O}$ with constant L_a ; that is, holding τ fixed, $\|\ddot{p}(x) - \ddot{p}(x')\| \leq L_a \|x - x'\|$.

(A4) **Secondary-torque consistency defect.** $\|JM^{-1}N_\tau\| \leq \kappa_{\text{sec}}$ and $\|\tau_{\text{sec}}\| \leq T_{\text{sec}}$ on $\mathcal{O}$. Exact dynamic consistency is $\kappa_{\text{sec}} = 0$.

(A5) **Bounded model errors.** The force-model error satisfies $\|\delta_F\| \leq \bar{\delta}_F$ and the torque-realization error $\|\delta_\tau\| \leq \bar{\delta}_\tau$, the latter being the same bound already used for tightening in (T1)–(T2).

B. The bound

Proposition 1 (A priori (T3) satisfaction): Let the manager hold its correction over $[t_k, t_{k+1}]$ and let $\bar{a}_k$ be the interval-averaged request of Corollary 1. Under (A1)–(A5), if

$$\Delta t (\eta_{\text{disc}} + \eta_{\text{hold}} + \eta_{\text{sec}} + L_F \bar{\delta}_F + L_\tau \bar{\delta}_\tau) \leq \epsilon_v, \quad (7)$$

with the five terms given by

$$\begin{aligned} \eta_{\text{disc}} &= L_a V_x T_f, & \eta_{\text{hold}} &= \sup_t \|a_f(t) - \bar{a}_k\|, \\ \eta_{\text{sec}} &= \kappa_{\text{sec}} T_{\text{sec}}, & L_F &= \bar{\lambda}, \\ L_\tau &= \bar{\sigma}_J / \underline{\mu}, \end{aligned} \quad (8)$$

then (T3) holds for that interval, and the conclusion of Theorem 1 follows without measuring the successor defect.

Proof: Write the realized task acceleration as $\ddot{p}(t) = \bar{a}_k + e(t)$. Since the abstract successor is $F_v(\Pi(x_k), \bar{a}_k) = v_k + \Delta t \bar{a}_k$ and the physical successor is $v_k + \int_{t_k}^{t_{k+1}} \ddot{p}(t) dt$, the (T3) defect is exactly $\|\int_{t_k}^{t_{k+1}} e(t) dt\| \leq \Delta t \sup_t \|e(t)\|$. It remains to bound $\|e(t)\|$ by the bracketed sum.

By (A1) the realization law (2) gives $\ddot{p} = a$ exactly when the model is exact, the torque is unclipped — guaranteed here by (T1)–(T2) — and the map is evaluated at the current state. Each bracketed term is one departure from that identity, and the departures are additive in acceleration:

Discretization. Within a fast period the realization map is held at its value at the sample instant while the state moves by at most $V_x T_f$ by (A2); (A3) converts this to an acceleration error $L_a V_x T_f$.

Hold. The manager predicts with the single value $\bar{a}_k$ while the fast loop applies $a_f(t)$; the instantaneous discrepancy is $\|a_f(t) - \bar{a}_k\|$ by definition. (Its integral vanishes by the definition of $\bar{a}_k$, so this term may be replaced by zero when the QP is posed on the interval average; it is retained for implementations that predict with $a_f(t_k)$.)

Secondary torque. The secondary term contributes $JM^{-1}N_\tau \tau_{\text{sec}}$ to task acceleration, which is zero under exact dynamic consistency and at most $\kappa_{\text{sec}} T_{\text{sec}}$ under (A4).

Force model. An uncompensated force error δ_F enters task acceleration through $\Lambda^{-1} = JM^{-1}J^\top$, whose norm is at most $\bar{\lambda}$ by (A1), giving $\bar{\lambda} \bar{\delta}_F$.

Torque. A torque error δ_τ enters through JM^{-1} , and $\|JM^{-1}\| \leq \sigma_{\max}(J) / \lambda_{\min}(M) \leq \bar{\sigma}_J / \underline{\mu}$ by (A1), giving $(\bar{\sigma}_J / \underline{\mu}) \bar{\delta}_\tau$.

Summing and multiplying by Δt yields (7). ■

C. Reading the bound

Three properties are worth noting.

First, (7) is *sufficient, not necessary*: it bounds each source by its worst case over $\mathcal{O}$ and adds them, so it is conservative by construction. The measured defect of Section VII-D and the a priori bound (7) are separate lines of evidence.

Second, the bound is *actionable in design*. Every term is a quantity a designer controls: η_{disc} falls linearly with the fast period, η_{hold} vanishes if the QP is posed on the interval average, η_{sec} vanishes under exact dynamic consistency, and the last two scale linearly with the declared force- and torque-error boxes. The manager period enters as an overall factor, which is the formal counterpart of the empirical observation that the slower rate is adequate only over a restricted operating region.

Third, it *closes the loop with (T1)*. The same $\bar{\delta}_r$ that tightens the actuator box in (T2) appears in the last term, so a single declared torque-error budget governs both the actuator margin and the certificate margin, tuned together.

Establishing the constants L_a , V_x , κ_{sec} , $\bar{\lambda}$, $\bar{\sigma}_J$ and $\underline{\mu}$ for a full rigid-body robot over a stated operating region is the identification task that connects the audited interface reported here to a deployment-time guarantee.

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